settings/parameters that optimize a process, but we do not know how the different settings influence and control the result of the process.

Further, trying out a particular setting and observing the result it has on the target process (the objective function) is time-consuming, expensive, or costly in some other sense.

> **IMPORTANT** **The act of "trying out a particular setting," that is, evaluating the value of the objective function at some input, is called making a query, or querying the objective function.**

The entire procedure is summarized in Figure 1.2.

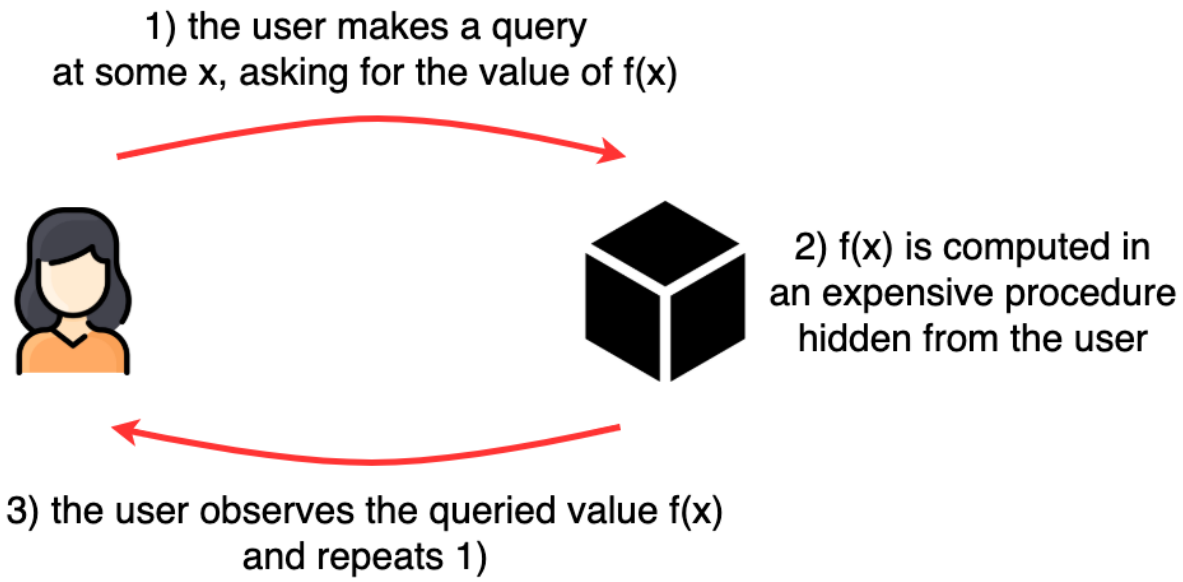


Figure 1.2 The framework of a black-box optimization problem. We repeatedly query the function values at various locations to find the global optimum.

1.1.3 Other real-world examples of expensive black-box optimization problems

Now, let's consider a few examples in the real world that fall into the category of expensive, black-box optimization problems. We will see that such problems are common in the real world, where we again have a function that we'd like to optimize, but evaluating it can only be done for a small number of times, so we'd like to find a way to intelligently choose where the evaluate the function.

The first example is drug discovery, the process in which scientists and chemists identify compounds with desirable chemical properties that may be synthesized into drugs. As you can imagine, the experimental process is quite involved and costs a lot of money. Another factor that makes this drug discovery task daunting is the decreasing trend in the productivity of drug discovery R&D that has been robustly observed in recent years. This phenomenon is known as *Eroom's law*—a reverse of *Moore's law*—which roughly states that the number of new drugs

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